Sahaj Bhadja

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Summary

Software Engineer with 2+ years of experience building **Java/Python applications**, **object-oriented systems**, automated unit/integration tests, and **deployment workflows** for reliable production software and technical support.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Operating Systems, Software Engineering

Experience

USC Dornsife | *Software Engineer* | *Los Angeles, CA* **September 2025 – Present**
- Achieved 99%+ reliability across 5K+ monthly jobs through **validation, reconciliation, and error handling**, cloud storage, CI/CD releases, and production monitoring supporting dependable internal software workflows and technical operations.
- Tripled concurrent throughput across 1K+ AI jobs by restructuring **fault-tolerant microservices** with explicit state transitions and failure isolation, preserving recoverable background processing reliably under sustained production workloads.

Grad8 | *Software Engineer Intern* | *Los Angeles, CA* **June 2025 – August 2025**
- Reduced feedback turnaround by 60% through a **production Python platform** supporting 150+ instructors and 10K+ daily requests with REST APIs, AWS RDS, secure ingestion, and CI/CD workflows.
- Processed 1K+ files per batch using **multithreaded validation and fault isolation**, preserving responsive workflows while troubleshooting heterogeneous PDF, DOCX, and ZIP inputs under concurrent model execution reliably.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) | *India*
- Improved API response times by 30% through **object-oriented microservices**, indexing, caching, pagination, and filtered payloads while debugging integrations across Windows, Linux, and macOS production environments reliably at scale.
- Automated builds and 20+ unit/integration suites through **Jenkins, GitHub, PyTest, and Mockito**, generating failure reports and coverage summaries across Linux, Windows, and macOS with repeatable nightly workflows.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) | *India*
- Synchronized 2M+ records through an eight-service **resource-management platform**, defining modular services for authentication, customers, billing, allocation, permissions, and notifications with real-time updates across distributed locations and operations reliably.
- Reduced resource wastage by 42% through **automated testing and deployment**, implementing JWT access controls, WebSocket updates, Azure monitoring, Docker/Jenkins pipelines, and ARM templates across eight modular service domains.

LST Capitals | *Web Developer Intern* | *India* **November 2022 – January 2023**
- Delivered **sub-200ms service APIs** across 50K+ booking records, implementing pricing, validation, payment webhooks, Firebase synchronization, and database transactions while preventing conflicting reservations across concurrent customer workflows reliably.
- Improved cross-platform compatibility by 15% through **React Native and TypeScript interfaces**, integrating Android, iOS, and web booking workflows while reducing design errors by 25% through reusable components.

Skills

Languages: Java, Python, C++, SQL, TypeScript, JavaScript, Bash
Software Design & Integration: Object-Oriented Programming, Data Structures, Algorithms, System Design, Microservices, REST APIs, WebSockets, Spring Boot, FastAPI, Flask
Testing & Simulation: Unit Testing, Integration Testing, Automated Testing, Regression Testing, Simulation, PyTest, Mockito, Jest, Selenium, Debugging, Fault Isolation, Root-Cause Analysis
Data & Platforms: PostgreSQL, MySQL, MongoDB, Azure SQL, AWS RDS, Firebase, Data Modeling, Query Optimization
Delivery & Support: Git, GitHub, Jenkins, Docker, Kubernetes, CI/CD, AWS, Azure, Monitoring, Error Handling, Data Validation, Technical Documentation, Agile, Scrum

Projects

Warehouse Equipment Emulator - Design Concept
- Designed a **warehouse-equipment emulator** concept modeling conveyors, sensors, stations, commands, and failure states with deterministic scenarios, fault injection, event logs, and integration reports for repeatable software commissioning tests.

CPU-Scheduling Simulator | Live
- Built a **CPU-scheduling simulator** implementing FCFS, SJF, SRTF, Priority, and Round Robin algorithms with configurable workloads, timeline visualization, comparative metrics, and deterministic validation across scheduling scenarios reliably.

Chat2SQL
- Extended a **Java and TypeScript database workspace** supporting 10+ database systems, using Docker, Maven, Node.js, and Yarn for cross-platform schema exploration, query workflows, local debugging, and deployment.